Started on	Monday, 4 April 2022, 2:15 PM
State	Finished
Completed on	Wednesday, 6 April 2022, 4:02 PM
Time taken	2 days 1 hour
Marks	5.00/7.00
Grade	**0.71** out of 1.00 (**71**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

The symbol table of a program is as below:

symbol	address
A	X400B
B	X400A
C	X400C
X	X4004
Y	X4003
Z	X4005

The hexadecimal representation of a machine instruction is x2C09. The address of the instruction is x4001. Which one of the assembly instructions below corresponds to the machine instruction?

Select one:

- a. LD R6, #8 ✖
- b. LD, R6, R1, #1
- c. LD R6, B
- d. LD R6, A
- e. LD R6, C

The correct answer is: LD R6, A

Question **2**

Incorrect

Mark 0.00 out of 1.00

What is the value passed to the OS stack during the execution of the instruction "TRAP x21" based on the following program?

Program:

```
        .ORIG   x3000
        LD      R2, TERM
        LD      R3, ASCII
AGAIN   TRAP    x20
        ADD     R1, R2, R0
        BRz     EXIT
        OUT
        ADD     R0, R0, R3
        TRAP    x21
        BRnzp   AGAIN
TERM    .FILL xFFc9
ASCII   .FILL x0020
EXIT TRAP   x25  ; halt
        .END
```

Select one:

- a. x3008
- b. x3007 ✖
- c. x3003
- d. x3005
- e. x3006

The correct answer is: x3008

Question **3**

Correct

Mark 1.00 out of 1.00

What is the value of R6 after the execution of the second "JSR Pop" on the following program?

Program:

```
        .ORIG   x3000
        AND     R0, R0, #0
        LD      R6, EMPTY
        ADD     R1, R0, #0
        JSR     Push ;first push
        ADD     R1, R0, #5
        JSR     Push ;second push
        ADD     R1, R0, #2
        JSR     Pop ;first pop
        ADD     R1, R0, #3
        JSR     Push ;third push
        ADD     R1, R0, #4
        JSR     Push ;forth push
        JSR     Pop ;second pop
        ADD     R0, R1, #0
        BRnzp   EXIT
Push    STR     R1, R6, #0
        ADD     R6, R6, #-1
        RET
Pop         ADD     R6, R6, #1
        LDR     R1, R6, #0
        RET
EMPTY   .FILL x4000
EXIT TRAP   x25     ; Halt
        .END
```

Select one:

- a. x3FFD
- b. x4000
- c. x3FFE ✔
- d. x3FFF
- e. x3FFC

The correct answer is: x3FFE

Question **4**

Correct

Mark 1.00 out of 1.00

What is the machine code of the instruction "BRz OUTP"?

Program:

```
        .ORIG   x3000
        AND     R2, R2, #0      ; R2 is counter, initially 0
        LD      R3, PTR         ; R3 is pointer to characters
        GETC                    ; R0 gets character input
        LDR     R1, R3, #0      ; R1 gets first character
TEST    ADD     R4, R1, #-10    ; Test for end of line (ASCII xA)
        BRz  OUTP
        NOT     R1, R1
        ADD     R1, R1, R0
        NOT     R1, R1
        BRnp    GETCH
        ADD     R2, R2, #1
GETCH   ADD     R3, R3, #1
        LDR     R1, R3, #0
        BRnzp   TEST
OUTP    LD      R0, ASCII
        ADD     R0, R0, R2
        OUT
        HALT                    ; Halt machine
ASCII   .FILL x0030
PTR     .FILL x4000
        .END
```

Select one:

- a. x0FFB
- b. x0FF6
- c. x0403
- d. x0A01
- e. x0408 ✔

The correct answer is: x0408

Question **5**

Correct

Mark 1.00 out of 1.00

What is the address of the label (symbol) "VAL2" in the following program?

Program:

```
        .ORIG   x3000
        LD      R1, VAL1
        LD      R2, VAL2
        LD      R5, VAL3
        AND     R3, R3, #0
; The inner loop
;
AGAIN   ADD     R3, R3, R2
        ADD     R1, R1, #-1
        BRp  AGAIN
;
        HALT
;
VAL1    .BLKW   #2
VAL2    .BLKW   #3
HELLO   .STRINGZ "Hello world"
VAL3    .FILL x0003
        .END
```

Select one:

- a. x300D
- b. x3004
- c. x300A ✔
- d. x3019
- e. x3008

The correct answer is: x300A

Question **6**

Correct

Mark 1.00 out of 1.00

```
        .ORIG X3000
        ADD   R7, R6, #-5
        LEA   R6, Y
        BRz   B
A       LDR   R0, R2, #2
B       HALT
C       .BLKW       5
X       .FILL  2
Y       .FILL  1
Z       .FILL  2
        .END
```

For the program above, what is the machine code of instruction "LDR R0, R2, #2"?

Select one:

- a. xE082
- b. x6082 ✔
- c. x2042
- d. x2082
- e. x6042

The correct answer is: x6082

Question **7**

Correct

Mark 1.00 out of 1.00

What is the value of the memory location at "x3FFF" after the execution of the third "JSR Push" subroutine on the following program?

Program:

```
        .ORIG   x3000
        AND     R0, R0, #0
        LD      R6, EMPTY
        ADD     R1, R0, #0
        JSR     Push ;first push
        ADD     R1, R0, #5
        JSR     Push ;second push
        ADD     R1, R0, #2
        JSR     Pop ;first pop
        ADD     R1, R0, #3
        JSR     Push ;third push
        ADD     R1, R0, #4
        JSR     Push ;forth push
        JSR     Pop ;second pop
        ADD     R0, R1, #0
        BRnzp   EXIT
Push    STR     R1, R6, #0
        ADD     R6, R6, #-1
        RET
Pop         ADD     R6, R6, #1
        LDR     R1, R6, #0
        RET
EMPTY   .FILL x4000
EXIT TRAP   x25     ; Halt
        .END
```

Select one:

- a. x0005
- b. x0006
- c. x0002
- d. x0004
- e. x0003 ✔

The correct answer is:
x0003

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